Ans:- $S_b = 5$ kmph, $S_w = 1$ kmph.

going to a place means downstream dist travelling & returning to original place means upstream dist travel.

$\Rightarrow t_d + t_u = 1hr = \frac{d}{S_d} + \frac{d}{S_u} = \frac{d}{S_b+S_w} + \frac{d}{S_b-S_w}$

$\Rightarrow \frac{d}{5+1} + \frac{d}{5-1} = 1 \Rightarrow \frac{2d+3d}{12} = 1.$

$\Rightarrow 5d = 12km \Rightarrow \boxed{d = 2.4\ km}$

Ex:-20:- [B.Q:-23]:- A man can row @ 12kmph speed on still water to a certain upstream point & back to the joining point on a river which flows @ 3kmph. Find his Average speed for this total journey?

Ans:- $S_m = 12$kmph, $S_w = 3$kmph.

Going with S_d & returning with S_u.

$S_d = S_m + S_w = 12+3 = 15$ kmph, $S_u = S_m - S_w = 12-3 = 9$kmph

going with 15 kmph & returning with 9kmph means Covering equal distance twice with 2 individual Speeds

$\therefore A.S$ (Average Speed) $= \frac{2xy}{x+y} = \frac{2\times15\times9}{15+9}$

$A.S = \frac{270}{24} = \frac{45}{4} = \boxed{11.25 kmph}$

Ex:-21:- [B.Q:-25]:- A man can row 7.5kmph on still water & he finds that it takes him twice as long to row up as to row down the river. Find the rate of stream?

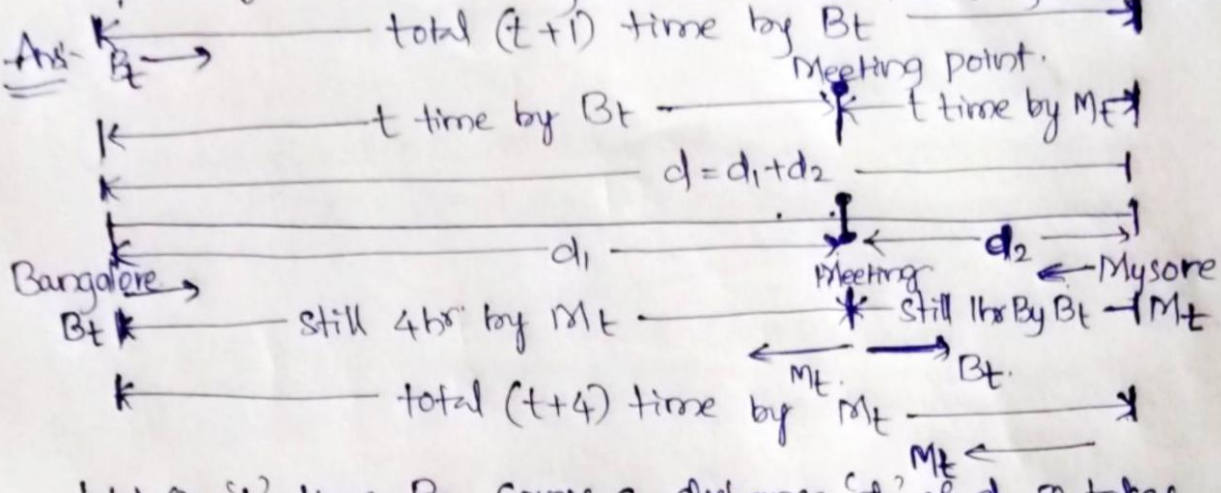
Ans:- $t_{upstream} = 2\ t_{downstream} \Rightarrow \frac{d}{7.5-x} = 2\frac{d}{7.5+x}.$

$7.5+x = 15-2x \Rightarrow \boxed{x = 2.5 kmph}$

(91)

Ex-22:- [B.Q:-26]:- Two trains starting @ same time, one from Bangalore to Mysore & other frm opposite direction arrive @ their destination 1 hr & 4 hr's after passing each other respectively. How much faster is one train from other?

Ans:-



↳ let frm 't' time Bt covers a distance 'd1' of d & takes 1 hr time to cover remaining distance 'd2' with speed $\frac{d}{(t+1)}$ after meeting train Mt. $\left[S_{Bt} = \frac{d}{t+1}\right]$

↳ Let frm 't' time Mt covers a distance of 'd2' of d & takes 4 hr's time to cover remaining distance 'd1' with speed $\left(\frac{d}{t+4}\right)$ after meeting train Bt. $\left[S_{Mt} = \frac{d}{t+4}\right]$

↳ from above graph $d = d_1 + d_2$

$$d = t\cdot\left(\frac{d}{t+1}\right) + t\left(\frac{d}{t+4}\right) \Rightarrow \frac{t}{t+1} + \frac{t}{t+4} = 1$$

$$\Rightarrow \frac{t(t+4)+t(t+1)}{(t+1)(t+4)} = 1 \Rightarrow t^2+4t+t^2+t = (t+1)(t+4)$$

$$\Rightarrow 2t^2+5t = t^2+1t+4t+4 \Rightarrow t^2 = 4 \Rightarrow t = \pm 2 \text{ but '+ve'}$$

$$t = +2.$$

$$\Rightarrow S_{Bt} = x\cdot S_{Mt} \Rightarrow \frac{d}{(t+1)} = x\cdot\frac{d}{(t+4)}$$

x = no. of times

$$\Rightarrow \frac{1}{3} = x\cdot\frac{1}{6} \Rightarrow \boxed{x = 2} \Rightarrow \boxed{S_{Bt} = 2\cdot S_{Mt}}$$

(92) (or)

Note:- When two objects starting @ same time from A & B towards each other & after crossing they take t_1 & t_2 times in reaching B & A respectively. Then S_1 & S_2 are

$$\frac{S_1}{S_2} = \sqrt{\frac{t_2}{t_1}}$$

In above Question we have $t_2 = 4$, $t_1 = 1$

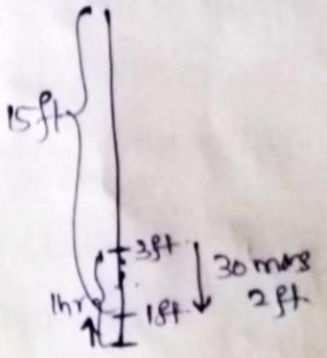
$$\Rightarrow \frac{S_1}{S_2} = \sqrt{\frac{4}{1}} = 2 \Rightarrow S_1 = 2S_2 \Rightarrow \boxed{S_{Bt} = 2S_{Mt}}$$

Ex-23:- [BQ:-27]:- A frog starts climbing 15 feet wall. Each hr it climbs 3 feet & rests 30 mins. During it slips 2 ft. How many hr's does it takes to reach the top?

Ans:- In 1hr climbs 3 feet & next in rest of 30 mins slips by 2 feets in covering 15 feet wall. So, that

In $(1hr + \frac{1}{2}hr) = (3ft - 2ft) = 1ft$.

like in 18 hr's 12 feets can be covered by frog $(12 \times 1\frac{1}{2} hr)$.



In the next 1 hour it jumps 3 ft & reaches the top means touches the 15th feet of wall. After reaching it slips in the next $\frac{1}{2}$ hr's it not to be considered.

Total time = $12 \times 1\frac{1}{2} + 1hr = \boxed{19 hr's}$

Ex:-24:- A, B & C running around a circular track of length 1200 m with respective speeds 9, 18 & 27 kmph. If they started @ same time from the same point & run in the same direction then.

(i) When will they meet for the first time @ any point?

(ii) " " " " " " @ starting point?

Ans:- (i) $S_A = 9 \times \frac{5}{18} = 2.5$ mps, $S_B = 18 \times \frac{5}{18} = 5$ mps, $S_C = 27 \times \frac{5}{18} = 7.5$ mps

$LCM\left(\frac{L}{a-b/b-a}, \frac{L}{b-c/c-b}\right)$

$= LCM\left(\frac{1200}{5-2.5}, \frac{1200}{7.5-5}\right) = LCM(480, 480) = 480$ sec

Meeting for the first time @ any point after 480 sec.

(or) 8 mins $\left(\frac{480}{60} = 8 \text{ m}\right)$

(ii) $LCM\left(\frac{L}{a}, \frac{L}{b}, \frac{L}{c}\right) = LCM\left(\frac{1200}{2.5}, \frac{1200}{5}, \frac{1200}{7.5}\right)$

$= LCM(480, 240, 160) = 480$ seconds

Meeting for the first time @ starting point is after 480 seconds (or) 8 mins.

↳ Means in this problem Meeting for the first time at any point is starting point only.